Case Study 1: A Statistical Analysis of Buchanan's False Votes in the 2000 Presidential Election in Palm Beach County

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The 2000 presidential election race between Democratic nominee Al Gore and Republican nominee George W. Bush was one of the closest in history with Bush leading with 400 votes in the final recount. Voters in Palm Beach county claimed to have accidentally voted for Reform Party candidate Pat Buchanan, which was supported by double stamping on butterfly ballots. Therefore, in this case study we attempt to answer if there was a significant increase of Buchanan votes in Palm Beach county compared to the other counties in Florida. We then, after figuring out if there is a discrepancy between the number of votes Buchanan and Bush received in Palm Beach compared to other counties, obtained an estimate of what votes were intended for Gore assuming that the excess in Buchanan votes were meant for Gore. To do all of this analysis, we use the votes cast for Bush in all of the other Florida counties to predict the number of votes Buchanan should have gotten in Palm Beach county.

Data Description & Exploratory Data Analysis

For our linear regression modeling, we will be utilizing the Slueth2 library that contains the dataset. This dataset contains information on the number of votes for Buchanan and Bush in all 67 counties in Florida obtained during the November 7, 2000 election. The voting information was also gathered from local voting machines. For each of the counties, there are corresponding variables that represent the voting turn out for Buchanan (listed as Buchanan2000) and Bush (Bush2000). The values assigned to the nominee variables help us to compare the number of votes cast for either candidate within each county.

Our research question primarily focuses on finding out whether the butterfly ballot confusion in Palm Beach County caused a significant increase in ballots cast for Buchanan that should have instead gone to Gore. To create a prediction interval for how many votes Buchanan should have received based on the overall Florida county ballot information, we must first remove Palm Beach County from our 67 observations, which leaves us with 66 observations or counties for our modeling. By removing Palm Beach, we are now able to create a Buchanan average vote prediction without interference from the confounding variable of confusing butterfly ballots that caused issues in this county. We did not remove any additional observations as there were no reported issues with their voting systems, nor were there unreported votes in the data set that caused NA values. Additionally, we found that the original data did not meet linearity, normality, or equal variance conditions so we transformed it using log(variable + 20).

Coefficients	Estimates	Standard Error	t-value	p-value
(Intercept)	-0.83043	0.28071	-2.95829	0.00433
log(Bush2000 + 20)	0.60017	0.02848	21.07261	0.00000

When analyzing the transformed data, we were focused on finding a prediction interval that would capture Buchanan's true vote count in Palm Beach County. To find this interval, we used Bush's votes across all 66 remaining counties as the explanatory variable (x) and the resulting Buchanan votes as our response variable (y). When fitting a linear regression model with the previously set variable assignments, we obtain a summary table that describes the relationship between log(Bush2000 + 20) and log(Buchanan2000 + 20) to be statistically significant due to a low p-value of < 2e-16, which is below the alpha of 0.05. They also share a positive relationship, so for every 1 vote given to Bush, Buchanan's average vote count increased by 0.60017 votes which is our slope. Additionally, when Bush has zero votes, Buchanan's average vote count would be -0.83043, which is the intercept of our data.

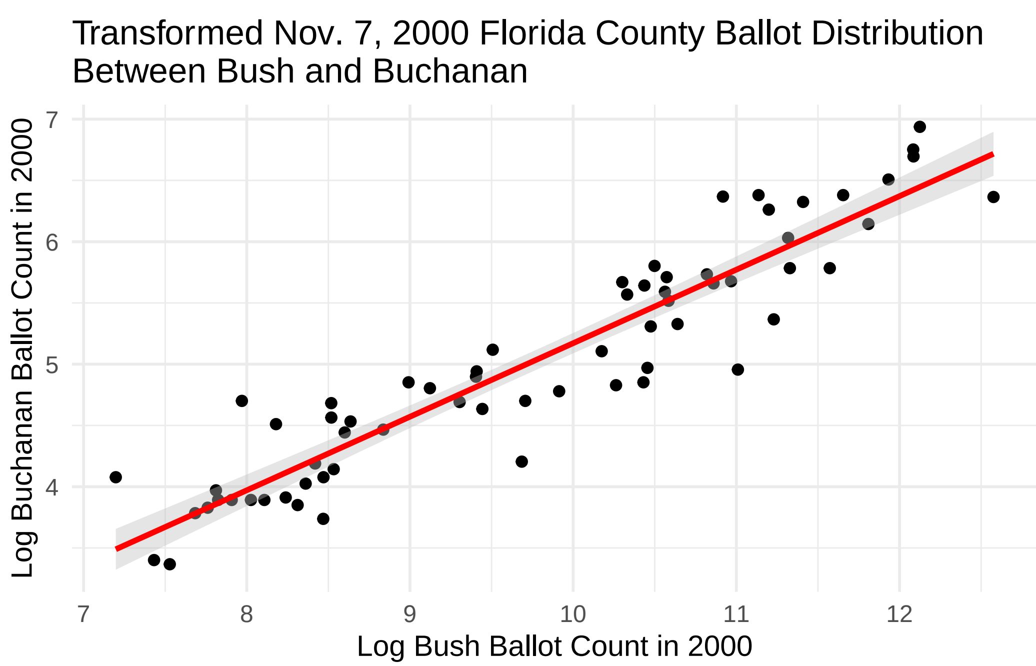


Figure 1: LOG transformed linear regression model for the distribution of Bush and Buchanan ballot votes in Florida counties.

Now we can fit our linear model to the original ballot data from the November 7, 2000 data to evaluate the fit (Figure 1).

Statistical Analysis & Results

Let $log(Buchanan2000 + 20)$ denote the base log 20 of Buchanan votes in each Florida county excluding Palm Beach county (measured in 1 vote per individual) and $log(Bush2000+20)$ denote the base log 20 of Bush votes in each Florida county excluding Palm Beach county (measured in 1 vote per individual). Our final linear regression model is

$$E[log(Buchanan2000 + 20)|log(Bush2000 + 20)] = \beta_0 + \beta_1 \left(log(Bush2000 + 20)\right).$$

. We then see in table 1 that our $beta_0$ is equal to -0.83043 and our $beta_1$ is equal to 0.60017. Their accompanying standard errors can similarly be viewed in table 1.

Coefficients	Estimates	Standard Error	t-value	p-value
(Intercept)	-0.83043	0.28071	-2.95829	0.00433
log(Bush2000 + 20)	0.60017	0.02848	21.07261	0.00000

Note:

Table 1

A log base 20 multiplicative change in the amount of votes Bush received is associated with a multiplicative change of $log + 20^{0.60017}$ in the median number of votes that Buchanan received. The results indicate that there was an unusual amount of Buchanan votes in Palm Beach county compared to other counties in Florida, meaning that some of the votes were most likely meant for Gore. The predicted average number of Buchanan votes was 563, while the true value of his votes was 3,407. Our prediction interval (286, 1111) does not include the true value of votes, providing further evidence that the number of Buchanan votes was unusually high in Palm Beach county. Based on our prediction interval and the data given, it is likely that the estimated number of votes that were intended for Gore was about 2,844.

Discussion

When we began this study our goal was to identify if there were discrepancies between the true ballot count that Buchanan received in Palm Beach county in 2000 and the predicted interval for his average vote count across the remaining 66 counties. From our analysis there is statistical evidence to support that there was a voting discrepancy in Palm Beach for Buchanan. His predicted average vote interval was from 286 to 1,111 votes. His true vote count from Palm Beach was 3,407 which lies outside of the the predicted 95% prediction interval generated from the 66 Florida counties. Moreover, his predicted average voting count from our model was 563 votes in Palm Beach. This means that there was a 2,844 ballot difference which indicates that these votes were potentially meant to go to Gore during the election. Therefore, we are concerned that the butterfly ballots used in Palm Beach county did cause voting interference and lost Gore the election. Based on this study we advise that counties like Palm Beach reconsider their ballot formatting for future elections so that constituents are not accidentally casting a vote for a candidate that they did not intend to support. However, within our study we only accounted for counted butterfly ballot interference and did not consider other potential issues such as ballots that were tossed out due to double stamping. In future studies having information about the ballots that were left uncounted could help inform the statistical model.

R Appendix

```r
election_wo_pb_lm <- lm(Buchanan2000 ~ Bush2000, data = election_wo_pb)
summary(election_wo_pb_lm)
```

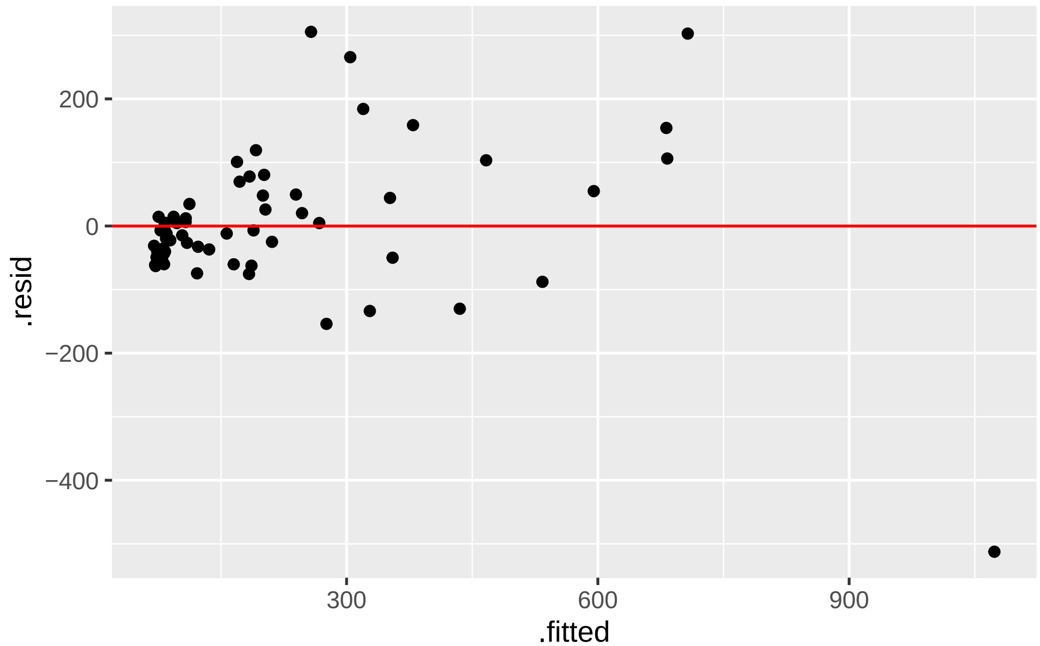
```
Call:
lm(formula = Buchanan2000 ~ Bush2000, data = election_wo_pb)

Residuals:
    Min      1Q  Median      3Q     Max 
-512.43  -47.97  -17.09   41.78  305.45 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 6.557e+01  1.733e+01   3.784 0.000343 ***
Bush2000    3.482e-03  2.501e-04  13.923  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 112.5 on 64 degrees of freedom
Multiple R-squared:  0.7518,    Adjusted R-squared:  0.7479 
F-statistic: 193.8 on 1 and 64 DF,  p-value: < 2.2e-16
```

```r
election.lm.table <- election_wo_pb_lm |> tidy()
election.lm.table |> kbl(col.names = c("Coefficients",
                                       "Estimates",
                                       "Standard Error",
                                       "t-value",
                                       "p-value"),
                         align = "c", booktabs = T, linesep="",
                         digits = c(5, 5, 5, 5)) |>
  kable_classic(full_width = F, latex_options = c("HOLD_position"))
```

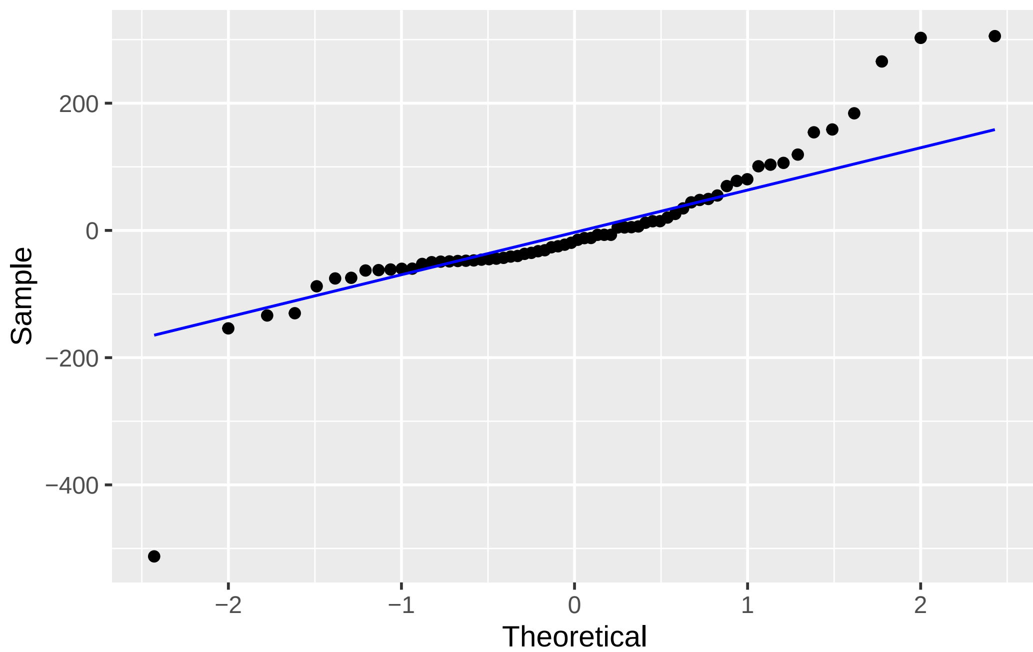
Coefficients	Estimates	Standard Error	t-value	p-value
(Intercept)	65.57350	17.33043	3.78372	0.00034
Bush2000	0.00348	0.00025	13.92256	0.00000

The two code chunks above display how we made the summary statistic table to get information for our fitted regression equation (assuming we did not need to do any transformations) from the original data set.

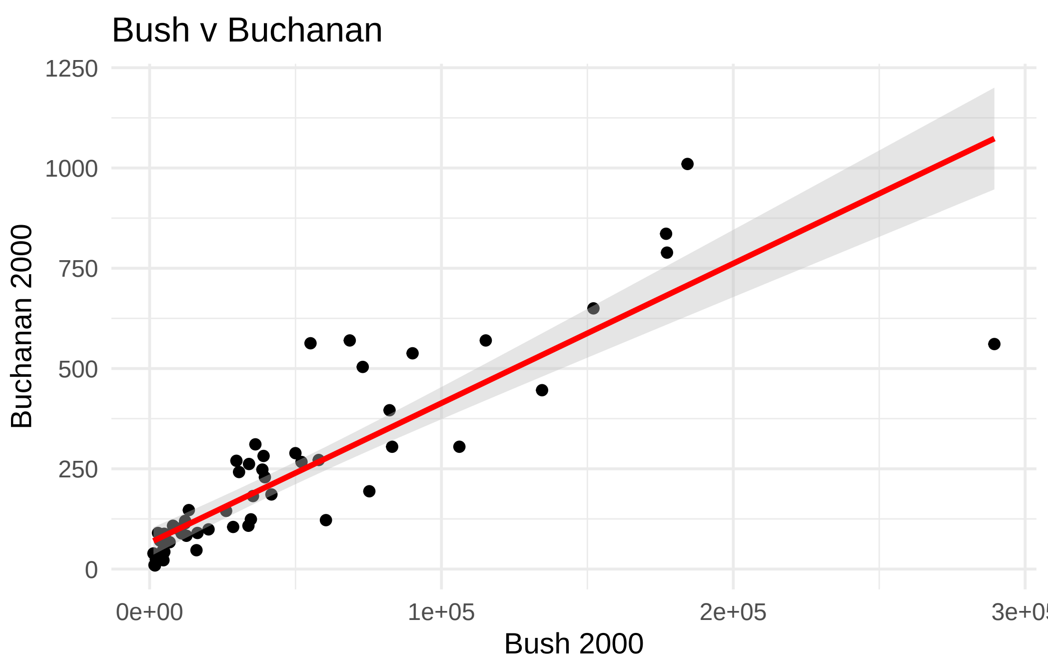
```r
election_wo_pb_lm |> augment() |> ggplot(aes(x = .fitted, y = .resid))+
  geom_point() +
  geom_hline(yintercept = 0, col = "red")
```




```r
election_wo_pb_lm |> augment() |> ggplot(aes(sample = .resid)) +
  geom_qq() +
  geom_qq_line(col = "blue") +
  xlab("Theoretical") + ylab("Sample")
```

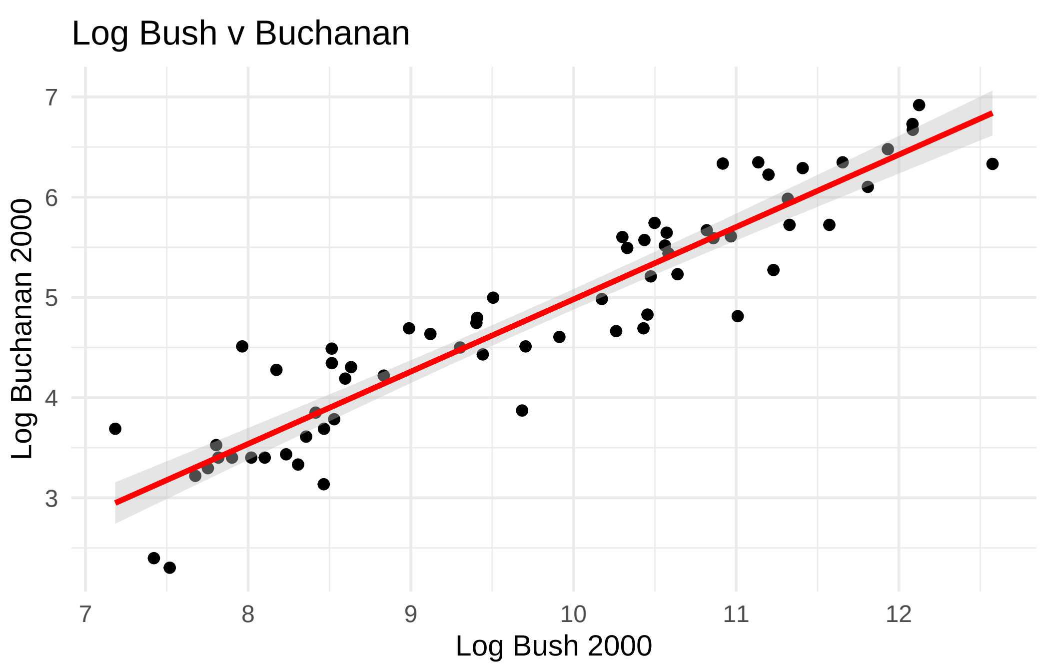



```r
ggplot(election_wo_pb, aes(x = Bush2000, y = Buchanan2000)) +
   geom_point() +
   geom_smooth(method = "lm", color = "red", fill = "grey", level = 0.95) +  #
   labs(
     title = "Bush v Buchanan",
     x = "Bush 2000",
     y = "Buchanan 2000")+
  theme_minimal()
```

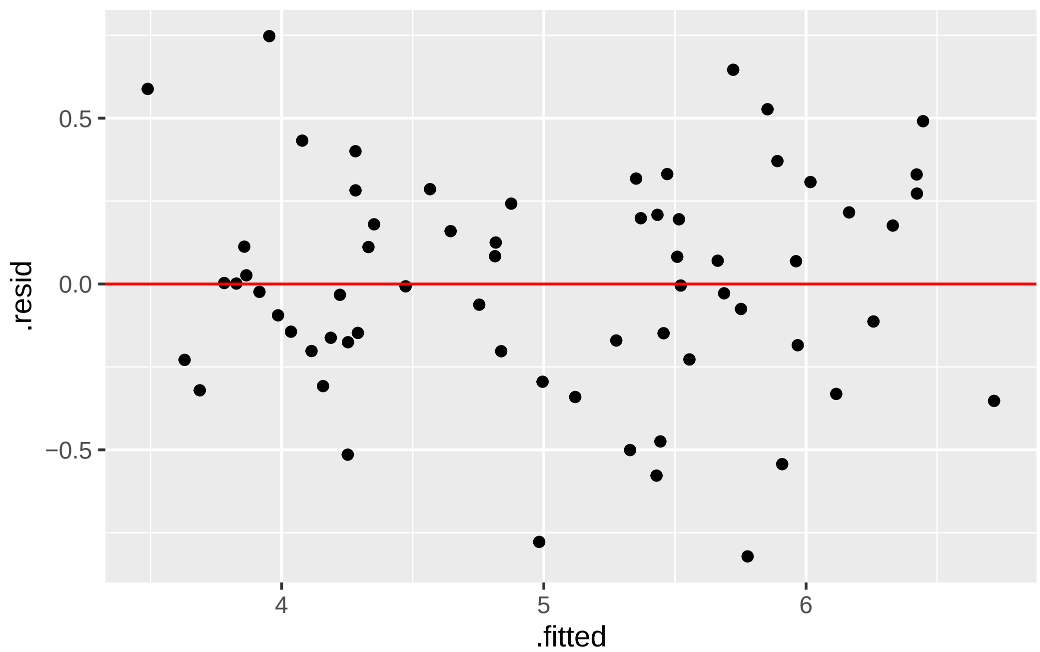


The three diagnostic plots above show how several conditions for a regression model were not met, including equal variance, linearity, normality. It was after seeing these plots that we decided to do a transformation on the data set to meet all the conditions.

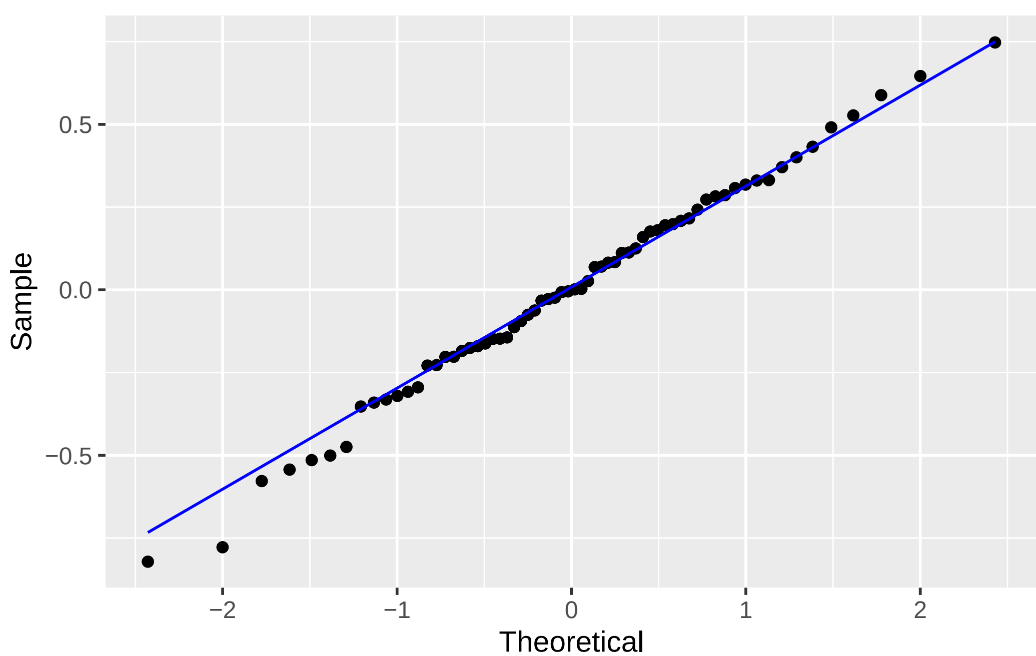
```
election_wo_pb <- election_wo_pb |> mutate(bush_log_one = log(Bush2000 + 1),
 buchanan_log_one = log(Buchanan2000 + 1))

ggplot(election_wo_pb, aes(x = bush_log_one, y = buchanan_log_one)) +
   geom_point() +
   geom_smooth(method = "lm", color = "red", fill = "grey", level = 0.95) +  #
   labs(
     title = "Log Bush v Buchanan",
     x = "Log Bush 2000",
     y = "Log Buchanan 2000",
) + theme_minimal()
```




```
ll_election_wo_pb_lm |> augment() |> ggplot(aes(x = .fitted, y = .resid))+
  geom_point() +
  geom_hline(yintercept = 0, col = "red")
```




```
ll_election_wo_pb_lm |> augment() |> ggplot(aes(sample = .resid)) +
  geom_qq() +
  geom_qq_line(col = "blue") +
  xlab("Theoretical") + ylab("Sample")
```



We then re-run the same three diagnostic plots and see that after transforming the data, the previous conditions mentioned are now nearly or fully met, which then allowed us to continue on with our data analysis.

```
new_election <- data.frame(Bush2000 = 152846)

prediction_interval <- election_wo_pb_lm |> augment(newdata = new_election,
                             interval = "prediction",
                             conf.level = 0.95)

kable(prediction_interval)
```

Bush2000	.fitted	.lower	.upper
152846	597.7677	364.709	830.8264

This was our prediction interval before the transformation was done on the dataset.

```
new_election <- data.frame(Bush2000 = 152846)

ll_prediction_interval <- ll_election_wo_pb_lm |> augment(newdata = new_election,
                          interval = "prediction",
                          conf.level = 0.95) |>
  select(c(".fitted", ".lower", ".upper")) |> exp()

kable(ll_prediction_interval)
```

.fitted	.lower	.upper
563.4241	285.6532	1111.301

This is our official prediction interval, which was done after the transformation.